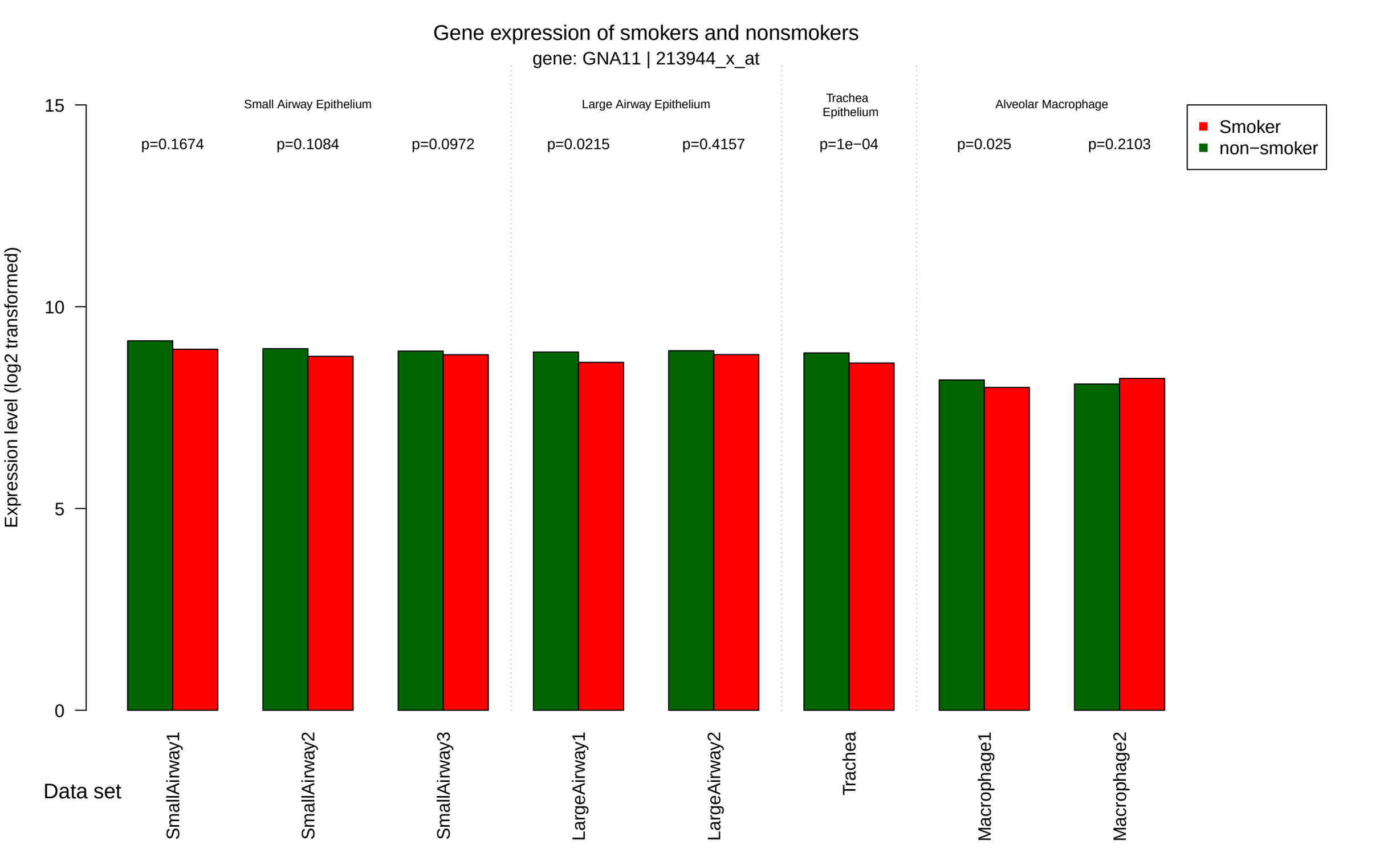
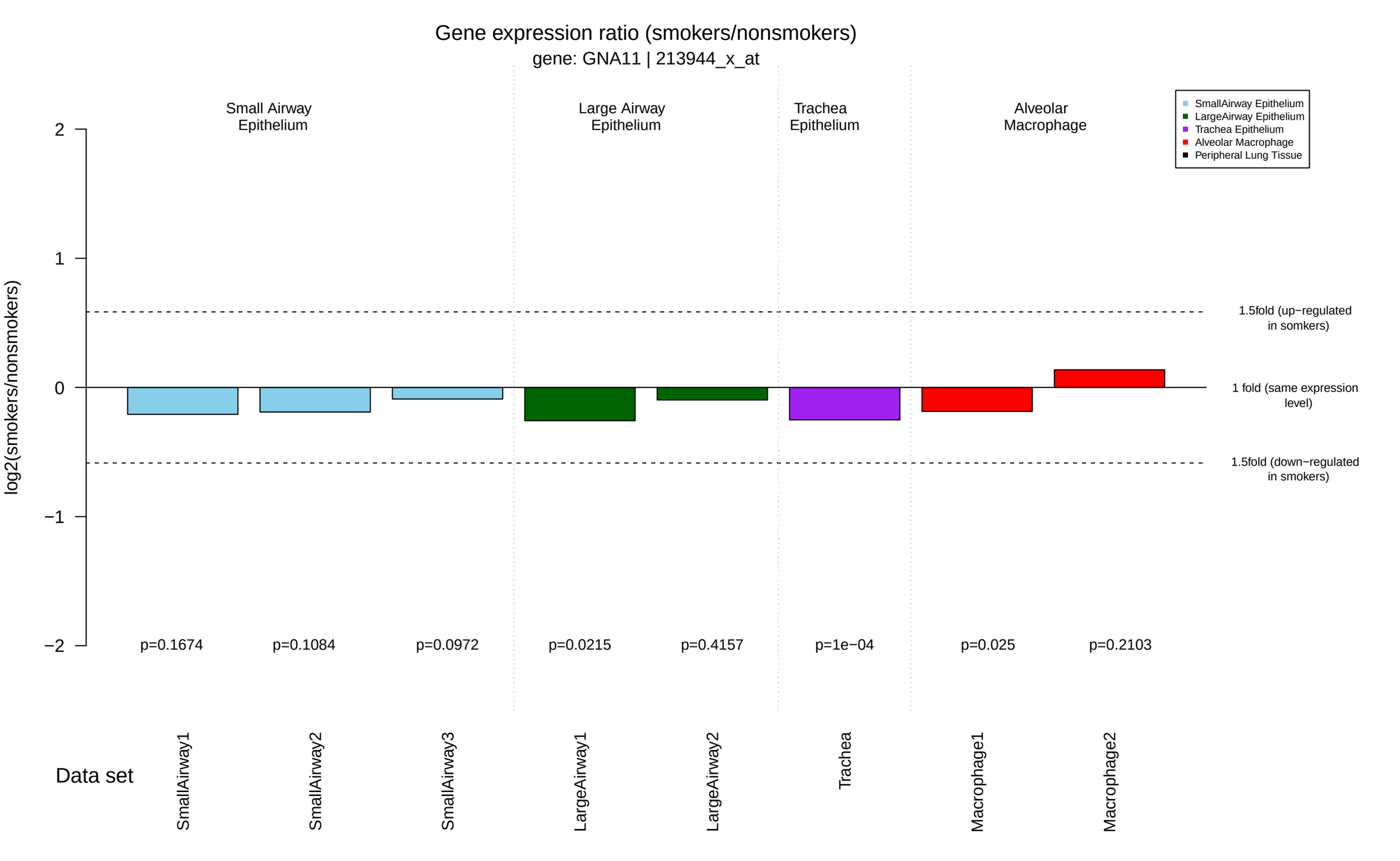
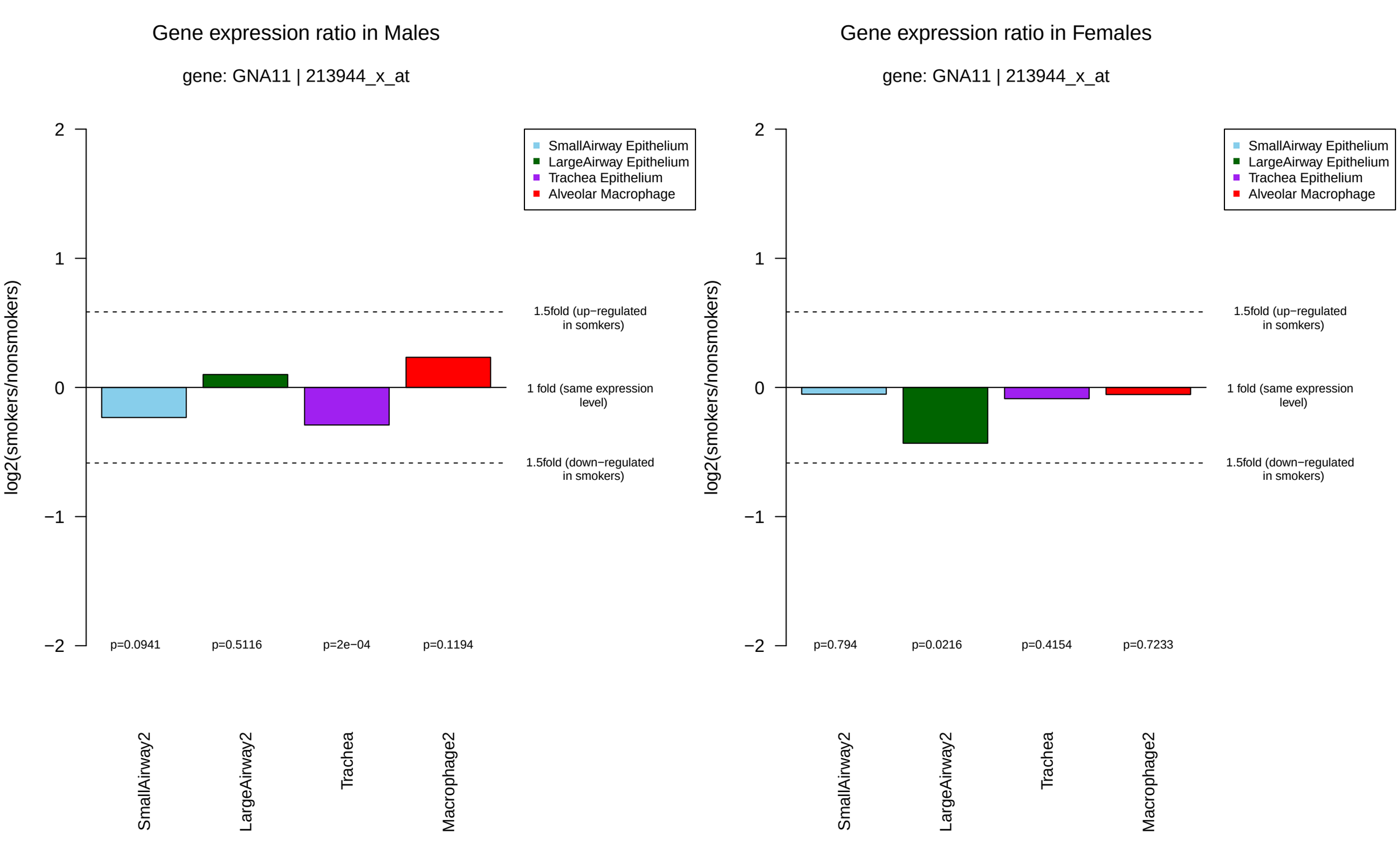
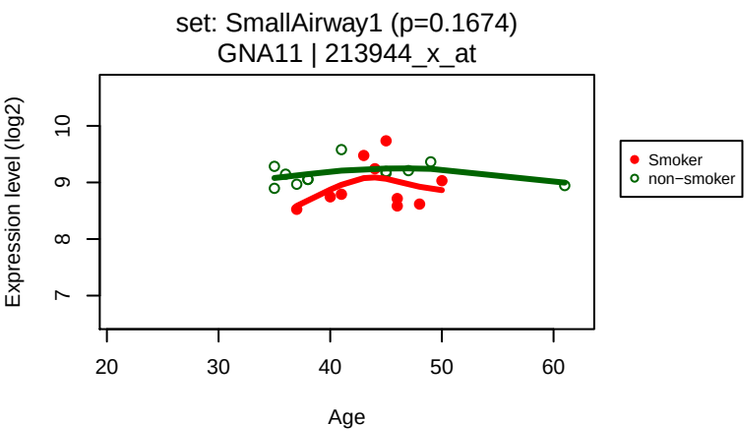
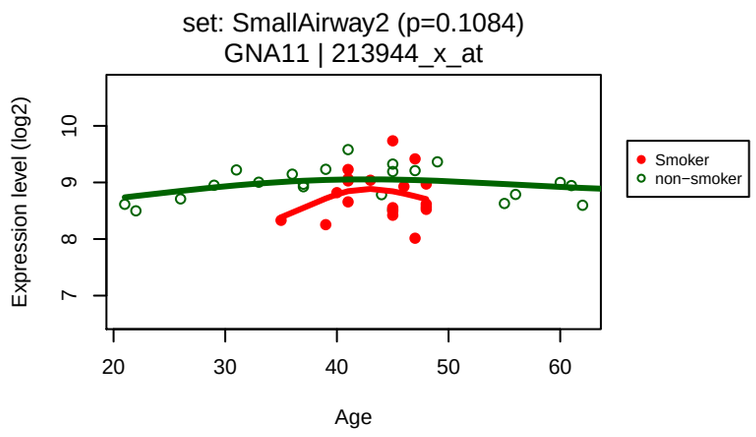
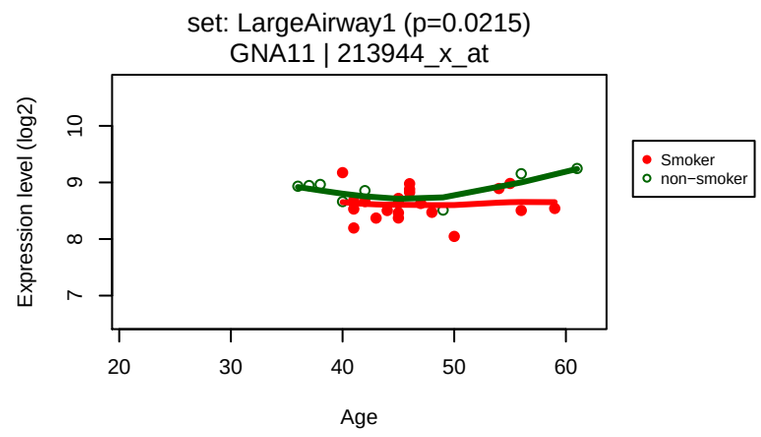

Gene expression of smokers and nonsmokers
gene: GNA11 | 213944_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1674
p=0.1084
p=0.0972
p=0.0215
p=0.4157
p=1e−04
p=0.025
p=0.2103
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

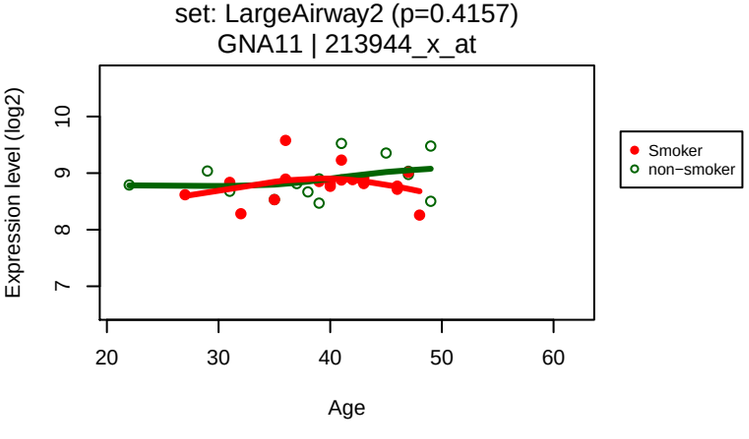

Gene expression ratio (smokers/nonsmokers)
gene: GNA11 | 213944_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1674
p=0.1084
p=0.0972
p=0.0215
p=0.4157
p=1e−04
p=0.025
p=0.2103
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

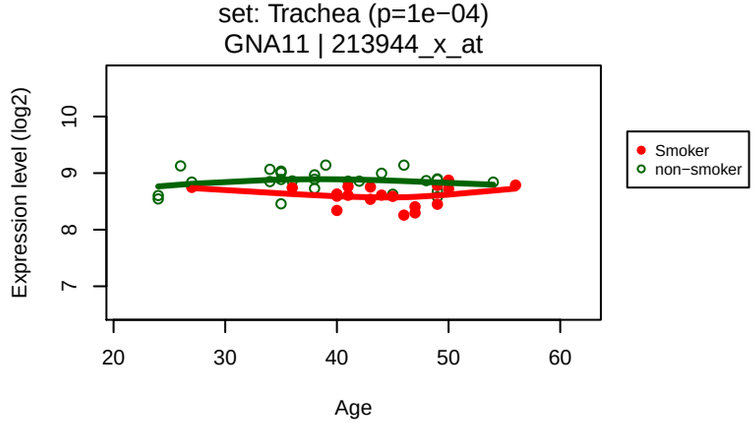

Gene expression ratio in Males
gene: GNA11 | 213944_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0941
p=0.5116
p=2e−04
p=0.1194
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA11 | 213944_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.794
p=0.0216
p=0.4154
p=0.7233
SmallAirway2
LargeAirway2
Trachea
Macrophage2

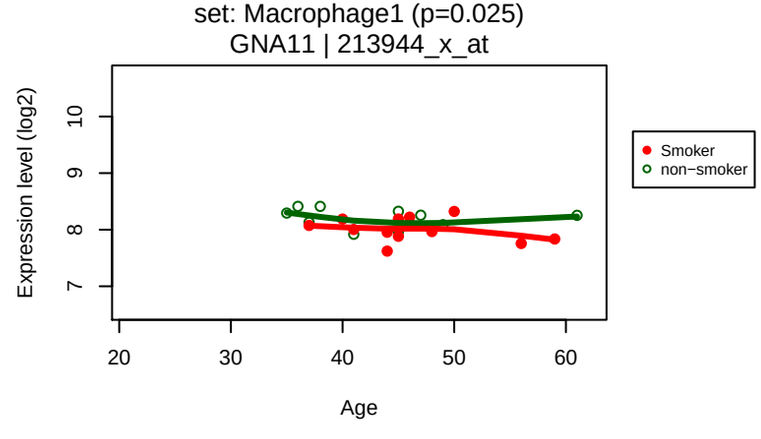

set: SmallAirway1 (p=0.1674)
GNA11 | 213944_x_at
Expression level (log2)
Age
Smoker
non−smoker

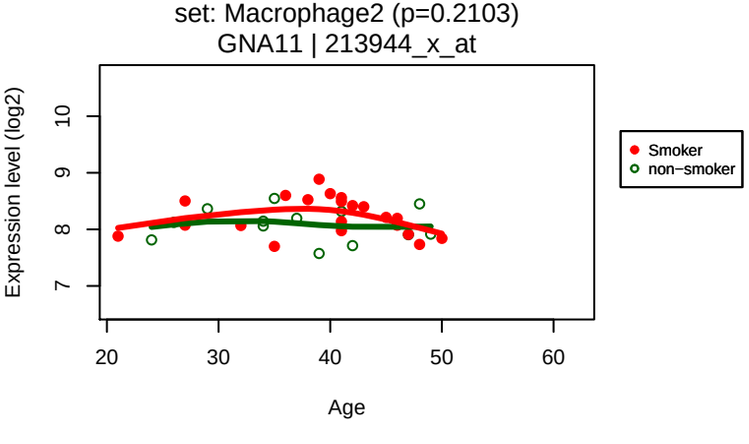

set: SmallAirway2 (p=0.1084)
GNA11 | 213944_x_at
Expression level (log2)
Age
Smoker
non−smoker

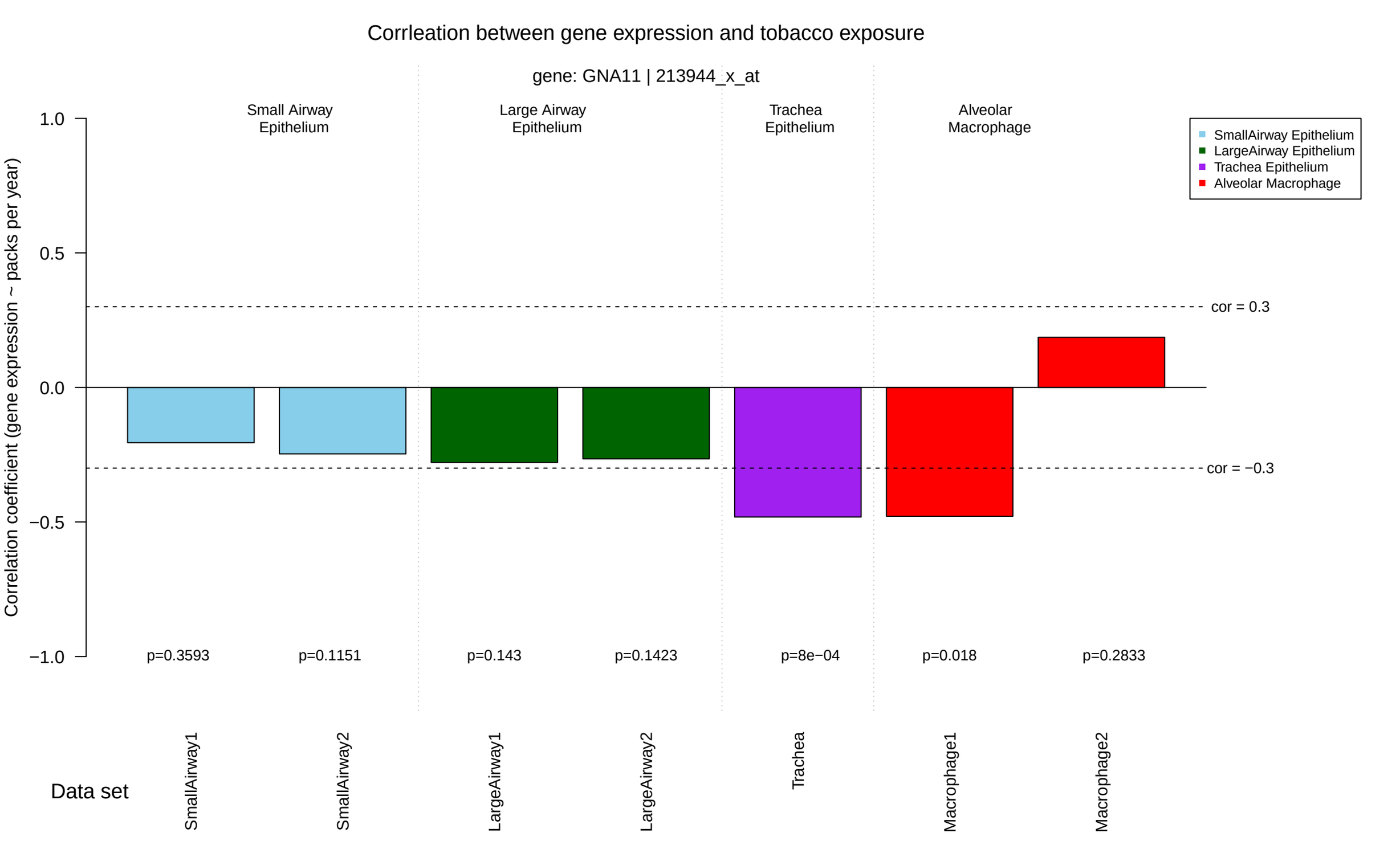

set: LargeAirway1 (p=0.0215)
GNA11 | 213944_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4157)
GNA11 | 213944_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
GNA11 | 213944_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.025)
GNA11 | 213944_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2103)
GNA11 | 213944_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA11 | 213944_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3593
p=0.1151
p=0.143
p=0.1423
p=8e−04
p=0.018
p=0.2833
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2